Group II Row IV: the BFKL term as the coefficient of $\log \vee$

Colour code: ~~red~~ = removed, blue = new. DGLAP, running coupling, finite and JIMWLK are as in `RowIV_derivation.pdf`. This note replaces only the BFKL argument (Steps 4–5 there).

1 The logic in one line

Whatever splitting, prescription or subtraction is used, the BFKL term of Row IV is fixed by one requirement: it is the part of Row IV that grows like $\log \vee$. So its coefficient is

$$\boxed{\text{coefficient of } \log\frac{\vee}{k^+} \;=\; \lim_{\vee\to\infty} \frac{d\,(\text{Row IV})}{d\log\vee}} \tag{1.1}$$

In the same way the JIMWLK coefficient is $\lim_{\Lambda\to 0}\frac{d\,(\text{Row IV})}{d\log(1/\Lambda)}$. These derivatives are computed below directly from your three lines. Nothing is split, added or subtracted, so the answer cannot depend on a convention.

2 Notation and starting point

- $d_\perp = 2-\epsilon$, $a \equiv k^+/\vee$, $X \equiv y'-y$, $\hat\mu^2 = \mu^2_{\overline{MS}} e^{2\gamma_E}/4$, $c \equiv \frac{g^2 N_c}{4\pi^2}$, $\mathcal{K}(r) \equiv r/r^2$.
- $\mathcal{B} \equiv \mathcal{B}_0 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ln(X^2\hat\mu^2)$ (exact).
- $F(\xi) \equiv \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[\mathcal{B}]$ is the LO with $\mathcal{B}$ *inside* the integral and phase $e^{-ik\cdot X/\xi}$. In particular $F(1) = \frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}]$.
- $F^{\mathrm{lp}}(\xi) \equiv \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{\mathrm{lp}}$.

Your three lines, with your p^+ prescription ($p^+ = k^+\frac{1-\xi}{\xi}$, $\frac{d\xi}{\xi^2}\frac{\xi}{1-\xi} = \frac{dp^+}{p^+}$):

$$\begin{aligned}
&(\mathrm{L1}) = c\int_a^1 \frac{d\xi}{\xi^2}\Big[\xi(1-\xi) + \frac{1-\xi}{\xi}\Big]F(\xi) \;+\; c\int_a^1 d\xi\,\frac{F(\xi)-F(1)}{\xi(1-\xi)},\\
&(\mathrm{L2}) = c\Big[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+}-1\Big)\Big]F(1), \qquad (\mathrm{L3}) = c\int_a^{\frac{k^+}{k^++\Lambda}} d\xi\,\frac{1-\xi}{\xi}\,F^{\mathrm{lp}}(\xi).
\end{aligned} \tag{2.1}$$

$\vee$ enters only through a, in the lower limits of (L1), (L3) and in the explicit log of (L2). Λ enters only in (L2) and in the upper limit of (L3).

3 Step 1: $d/d\log\vee$, term by term (exact)

Use $\frac{da}{d\log\vee} = -a$, so $\frac{d}{d\log\vee}\int_a^1 g(\xi)\,d\xi = a\,g(a)$. Also $\frac{d}{d\log\vee}\log(\frac{\vee}{k^+}-1) = \frac{1}{1-a}$.

piece	$\frac{d}{d\log\vee}$ of it (divided by c)	comes from
(L1), $\xi(1-\xi)$	$a\cdot\frac{a(1-a)}{a^2}F(a) = (1-a)F(a)$	$p^+\simeq\vee$
(L1), $\frac{1-\xi}{\xi}$	$a\cdot\frac{(1-a)/a}{a^2}F(a) = \dfrac{1-a}{a^2}F(a)$	$p^+\simeq\vee$
(L1), plus: $F(\xi)$ part	$a\cdot\frac{F(a)}{a(1-a)} = \dfrac{F(a)}{1-a}$	$p^+\simeq\vee$
(L1), plus: $-F(1)$ part	$-\dfrac{F(1)}{1-a}$	subtraction
(L2), $\log(\frac{\vee}{k^+}-1)$	$+\dfrac{F(1)}{1-a}$	explicit log
(L3)	$a\cdot\frac{1-a}{a}F^{\mathrm{lp}}(a) = (1-a)F^{\mathrm{lp}}(a)$	$p^+\simeq\vee$

The two blue entries cancel exactly. Hence, with no approximation,

$$\frac{d\,(\text{Row IV})}{d\log\vee} = c\,F(a)\Big[\frac{1-a}{a^2} + (1-a) + \frac{1}{1-a}\Big] + c\,(1-a)F^{\rm lp}(a). \tag{3.1}$$

Conclusion of Step 1. The derivative does not contain $F(1) = \frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]$ at all. So the cross section has *no* $\log\vee$ multiplying the LO Wilson-line structure:

- The $\log(\frac{\vee}{k^+}-1)F(1)$ of (L2) is not a large log of Row IV.
- It is cancelled identically by the $-F(1)$ inside your plus prescription, since $c\int_a^1 \frac{d\xi}{\xi(1-\xi)}\,F(1) = c\log(\frac{\vee}{k^+}-1)F(1)$.
- Equivalently, (L1)+(L2) is the unsubtracted integral $\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}$, whose $\vee$-dependence sits only at its endpoint $p^+ = \vee - k^+$.

~~BFKL = $c\log\frac{\vee}{k^+}\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]$~~. This is why splitting $\log(\frac{\vee}{k^+}-1) = \log\frac{\vee}{k^+} + \log(1-\frac{k^+}{\vee})$ in (L2) cannot give your Wilson-line structure: that log is not the BFKL log.

4 Step 2: what $F(a)$ is at small a

$F(a)$ is the LO evaluated at the large momentum $\mathbf{k}/a$:

$$F(a) = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',y}\int d^2y'\,e^{-i\mathbf{k}\cdot(y'-y)/a}\,\mathcal{B}(y'-y)\,\Phi_y(y'), \tag{4.1}$$

where $\Phi_y(y') \equiv \mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,[U^{e'c}(y') - U^{e'c}(x')][U^{ec}(y) - U^{ce}(x)]\rho^{e'}(x')\rho^e(x)$ is smooth at $y'=y$.

1. The $-\frac{2}{\epsilon}$ and $\ln\hat\mu^2$ in $\mathcal{B}$ are constants. They multiply $\int d^2y' e^{-i\mathbf{k}\cdot(y'-y)/a} = (2\pi a)^2\delta^2(\mathbf{k}) = 0$ for $\mathbf{k}\neq 0$. (So no pole and no μ survive.)
2. $\nabla^2\ln X^2 = 4\pi\delta^2(X)$ gives $\int d^2y'\,e^{-i\mathbf{k}\cdot(y'-y)/a}\big[-\ln(y'-y)^2\big] = +\frac{4\pi a^2}{\mathbf{k}^2}$. At large $\mathbf{k}/a$ only $y'\to y$ contributes, so $\Phi_y(y')\to\Phi_y(y)$ (checked numerically in `smallxi.py`).
3. At $y'=y\equiv z$ the weight is $\Phi_z(z) = \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)[U(z)-U(x')][U(z)-U(x)]\rho\rho$.

Therefore

$$F(a) = \frac{4\pi a^2}{\mathbf{k}^2}\,\frac{d^3N_{\rm LO}}{d^3k}^{\,y}\,\big[1+O(a)\big], \tag{4.2}$$

$$\frac{d^3N_{\rm LO}}{d^3k}^{\,y} \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',z}\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\big[U^{e'c}(z)-U^{e'c}(x')\big]\big[U^{ec}(z)-U^{ce}(x)\big]\rho^{e'}(x')\rho^e(x). \tag{4.3}$$

In the same way $F^{\rm lp}(a) = O(a^2)$.

5 Step 3: the BFKL coefficient

Insert Step 2 into Eq. (3.1):

- $\frac{1-a}{a^2}F(a)\to\frac{4\pi}{\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{\,y}$;
- $(1-a)F(a)$, $\frac{F(a)}{1-a}$ and $(1-a)F^{\rm lp}(a)$ are $O(a^2)$.

$$\boxed{\frac{d\,(\text{Row IV})}{d\log\vee} = \frac{c\,4\pi}{\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{\,y} + O\Big(\frac{k^+}{\vee}\Big) = \frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{\,y} + O\Big(\frac{k^+}{\vee}\Big) = H_4 + O\Big(\frac{k^+}{\vee}\Big)} \tag{5.1}$$

The last step uses $\int_w e^{i\mathbf{k}\cdot w}\mathcal{K}^i(w-z) = \frac{2\pi i k^i}{\mathbf{k}^2}e^{i\mathbf{k}\cdot z}$, so

$$\int_{w,w'} e^{-i\mathbf{k}\cdot(w'-w)}\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z) = \frac{4\pi^2}{\mathbf{k}^2} \tag{5.2}$$

(finite, no $\log\mu$). Hence $\frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{\,y}$ is exactly your four-line term:

$$H_4 = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5k^+}\int_{x,x',z,w,w'} e^{-i\mathbf{k}\cdot(w'-w)}\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$$

$$\times \Big[U^{e'c}(x')U^{ce}(x) + \delta^{e'e} - U^{e'c}(z)U^{ce}(x) - U^{e'c}(x')U^{ec}(z)\Big]\rho^{e'}(x')\rho^{e}(x), \quad (5.3)$$

which is $[U^{e'c}(z) - U^{e'c}(x')][U^{ec}(z) - U^{ce}(x)]$ expanded. $U^{e'c}(z)U^{ec}(z) = \delta^{e'e}$ gives your $\rho\rho$ line, and the last two are your $-U(z)U(x)$ and $-U(z)U(x')$ lines.

Integrating Eq. (5.1) in $\log\vee$:

$$\boxed{\text{Row IV} = \log\frac{\vee}{k^+}\,H_4 + \big(\text{terms independent of }\vee\big) + O\Big(\frac{k^+}{\vee}\Big).} \quad (5.4)$$

Why this is the physics of BFKL. The whole coefficient comes from the $\frac{1-\xi}{\xi}\frac{1}{\xi^2} \sim \frac{1}{\xi^3}$ term, i.e. the $\frac{1}{\xi}$ pole of $\mathcal{P}_{gg}$. This is the small-x (BFKL) term, and $\xi \to 0$ means $p^+ \gg k^+$:

- a fast gluon is produced at z (LO^y, inclusive in its transverse momentum);
- the measured gluon, much softer in k^+, is emitted from it with the soft kernel $\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$;
- $\frac{1}{\xi^3}$ times ξ^2 (Step 2) leaves $\frac{d\xi}{\xi}$, i.e. one unit of $\log\vee$ per unit of rapidity of the fast gluon;
- the structure is symmetric under amplitude $\leftrightarrow$ conjugate: both gluon points sit at z.

The opposite end, $\xi \to 1$ ($p^+ \to \Lambda$, the $\frac{\xi}{1-\xi}$ pole), is the soft *emitted* gluon. It keeps the LO structure, and that is JIMWLK (Step 4).

6 Step 4: $d/d\log(1/\Lambda)$ gives JIMWLK (same method)

- (L1) does not depend on Λ.
- (L2) gives $c\,F(1)$.
- (L3): $\frac{d\xi_{\max}}{d\log(1/\Lambda)} = \xi_{\max}\frac{\Lambda}{k^+ + \Lambda}$ with $\xi_{\max} = \frac{k^+}{k^+ + \Lambda}$, and the integrand $\propto (1-\xi_{\max}) = \frac{\Lambda}{k^+ + \Lambda}$, so it is $O(\Lambda^2/k^{+2})$.

$$\frac{d\,(\text{Row IV})}{d\log(1/\Lambda)} = c\,\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}] + O\Big(\frac{\Lambda}{k^+}\Big) \quad\Rightarrow\quad \text{JIMWLK} = c\log\frac{k^+}{\Lambda}\frac{d^3N_{\rm LO}}{d^3k}\Big[-\frac{2}{\epsilon} - \ln(X^2\hat{\mu}^2)\Big]. \quad (6.1)$$

This agrees with what you have.

7 Step 5: why the other choice is inconsistent

Suppose BFKL were taken as $c\log\frac{\vee}{k^+}\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]$. Then the "rest" would be

$$\frac{d}{d\log\vee}\Big[\text{Row IV} - c\log\frac{\vee}{k^+}\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]\Big] = H_4 - c\,\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}] \neq 0. \quad (7.1)$$

The DGLAP + RC + finite part would still grow like $\log\vee$, which contradicts "finite". With $\log\frac{\vee}{k^+}H_4$ the rest has zero $\log\vee$ coefficient. Numerical check (`dlogvee.py`): a test F with $F(\xi) = \xi^2 G(\xi)$, $c\,G(0) \equiv H_4 = 1.7$, $F(1) \equiv \frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}] = 0.9$:

$a = k^+/\vee$	$\frac{d\,\text{Row IV}}{d\log\vee}$	$\frac{d}{d\log\vee}\big[\text{Row IV} - \log\frac{\vee}{k^+}H_4\big]$	$\frac{d}{d\log\vee}\big[\text{Row IV} - \log\frac{\vee}{k^+}c\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]\big]$
10^{-2}	1.675418	−0.024582	0.775418
10^{-3}	1.697504	−0.002496	0.797504
10^{-4}	1.699750	−0.000250	0.799750

- The middle column $\to 0$ like a.
- The last column $\to H_4 - F(1) = 0.8$, not 0.
- The identity (3.1) holds to 10^{-11} for an arbitrary smooth F.
- The (L2) and plus-subtraction contributions are $+0.33317$ and -0.33317.

8 Step 6: the rest is DGLAP + RC + finite (as you have)

Subtract JIMWLK and BFKL. What is left has no $\log\vee$ and no $\log\Lambda$, and it is the result you already accept:

$$\begin{aligned}\text{Row IV} = c\log\frac{k^+}{\Lambda}\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}] \;+\; \log\frac{\vee}{k^+}H_4 \\ -\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}(\xi)\frac{d^3N_{\xi\,\rm LO}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big]+\frac{8\pi N_c\,\xi(1-\xi)}{\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{\,y}\Big\} \qquad \text{(DGLAP)} \\ +\frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\rm LO}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big] \qquad \text{(RC)} \\ -\Big(1-\frac{k^+}{\vee}\Big)H_4+c\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\frac{d^3N_{\xi\,\rm LO}}{d^3k}^{\,\rm lp}\Big|_{d_\perp=2} \qquad \text{(finite).}\end{aligned} \tag{8.1}$$

- The blue term inside DGLAP is not a new physics term. It is the $\frac{1}{\xi}$-pole part of the integrand already counted in $\log\frac{\vee}{k^+}H_4$:

$$\int_a^1\frac{d\xi}{\xi^2}\frac{8\pi N_c\xi(1-\xi)}{\mathbf{k}^2}\frac{g^2}{8\pi^2}\frac{d^3N_{\rm LO}}{d^3k}^{\,y}=\Big[\log\frac{\vee}{k^+}-(1-a)\Big]H_4. \tag{8.2}$$

- It has no pole and no μ, so the DGLAP renormalization ($\frac{2}{\epsilon}$ into $D_{h/g}$, $\ln\frac{X^2\mu_F^2}{c_0^2}$ left) is unchanged.
- The three blue H_4 pieces add to zero, so the sum is still exactly your three lines.
- One can check that $\frac{d}{d\log\vee}$ of the DGLAP line plus the finite line vanishes as $a\to 0$, as required by Step 3.

Summary

term	defined as	result
JIMWLK	$\lim\frac{d\,\text{Row IV}}{d\log(1/\Lambda)}$, from (L2) only	$c\log\frac{k^+}{\Lambda}\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]$, LO structure
BFKL	$\lim\frac{d\,\text{Row IV}}{d\log\vee}$, from $\xi\to 0$ of the $\frac{1-\xi}{\xi}$ term	$\log\frac{\vee}{k^+}H_4$, structure $[U(z)-U(x')][U(z)-U(x)]$
$\log(\frac{\vee}{k^+}-1)$ of (L2)	cancels the $-F(1)$ of your plus prescription	not a log of the cross section